Predicting Sleep Rhythm Shifts with an Attention-Based Model

Alice Smith Bob Lee
Department of Electrical Engineering, Example University
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Abstract: This is a sample document for testing the English version of the report template. All data and conclusions here are made up. Disrupted sleep rhythms are common in high-latitude winters. We propose an attention-based method f_θ that uses continuous data from wearable devices to predict the sleep phase shift for the coming week.

Experiments show that the method has a lower mean absolute error than the baselines on a simulated dataset, and that it can explain the contribution of each feature.

Keywords: sleep rhythm; attention mechanism; time series forecasting

1 Introduction

The sleep rhythm is one of the most important physiological rhythms in the human body. Disruption affects cognition, mood, and metabolic health[1]. As wearable devices become common, continuous recording of heart rate, activity, and skin temperature has become practical, which provides a basis for personalized rhythm prediction. Long words such as electroencephalography and internationalization should not break the justified text layout badly.

The rest of this report is organized as follows. Section 2 describes the prediction model, Section 3 presents the results, and Section 4 concludes. The overall framework is shown in Figure 2-1, the core formula is given in Eq. (2-1), and the results are listed in Table 3-2.

1.1 Background

Traditional assessment relies on laboratory melatonin tests, which are expensive and hard to repeat over long periods. Methods based on wearable devices are cheaper, but their data are noisy and vary widely between people.

1.2 Contributions

The main contributions of this work are:

1. an attention model that combines several physiological signals;
2. a masking mechanism for missing data;
3. an evaluation on a simulated dataset.

2 Method

2.1 Overall Framework

The overall framework is shown in Figure 2-1. The input is preprocessed and encoded, the attention module extracts features, and the model then predicts the phase for the coming week.

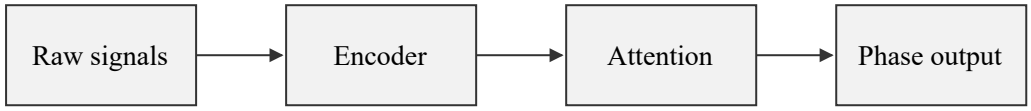


Figure 2-1. Overall framework of the proposed model

2.2 Attention Module

Given a query Q, keys K, and values V, scaled dot-product attention is computed as in Eq. (2-1):

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V \tag{2-1}$$

where d_k is the dimension of the keys. The loss function is defined as:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N} |\hat{y}_i - y_i| + \lambda \left\|\theta\right\|_2^2 \tag{2-2}$$

To simplify the derivation, here is an intermediate step that does not need a number:

$$\hat{y} = g(x; \theta)$$

2.2.1 Missing Data

Wearable data often have gaps. We use a mask matrix to mark the missing positions and exclude them when computing Eq. (2-1).

2.2.2 Implementation

The core code is in Listing 2-1:

Listing 2-1. model/attention.py

```
def attention(q, k, v, mask=None):
    score = q @ k.transpose(-2, -1) / k.size(-1) ** 0.5
    if mask is not None:
        score = score.masked_fill(mask == 0, -1e9)
    return score.softmax(dim=-1) @ v
```

3 Experiments

3.1 Dataset

The simulated dataset is summarized in Table 3-1.

Table 3-1. Summary of the dataset

Split	Subjects	Sampling rate	Duration
Training	120	1 Hz	90 days
Validation	30	1 Hz	90 days
Test	50	1 Hz	30 days

3.2 Comparison

Table 3-2 lists the prediction error of each method, and Figure 3-1 shows the error curves. The proposed method has the lowest error, as defined by Eq. (2-2).

Table 3-2. Prediction error of each method, where a lower MAE is better

Method	MAE (min)	RMSE (min)	Parameters
Linear regression	42.3	55.1	0.1 K
LSTM	31.7	40.2	85 K
Proposed	**24.6**	**31.8**	120 K

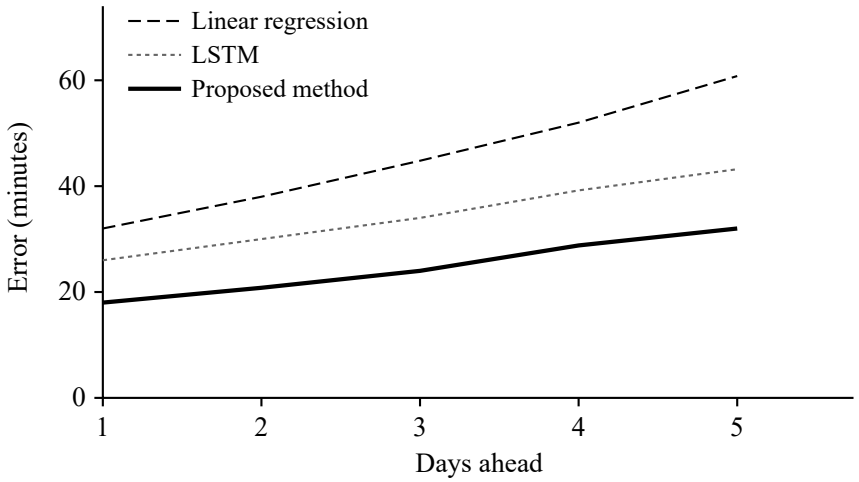
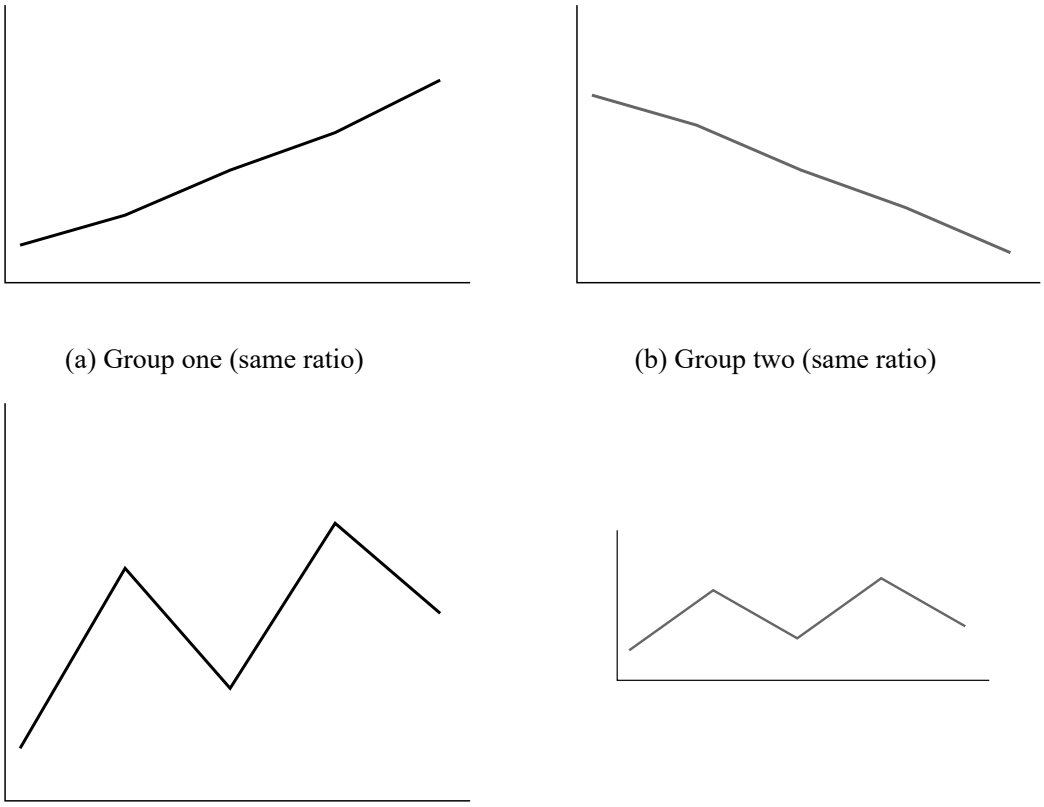


Figure 3-1. Error of each method as the horizon grows, where e_t is the error on day t

The errors of the four groups are shown in Figure 3-2, and Figure 3-2(c) has the largest variation. A figure group shares one number, and each panel has its own

caption.



(a) Group one (same ratio)

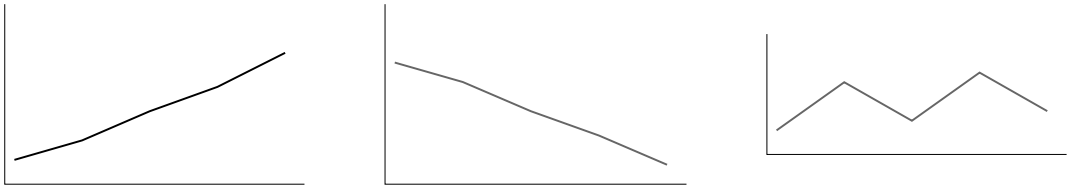
(b) Group two (same ratio)

(c) Group three (a taller image)

(d) Group four (a shorter image), with a longer caption and a formula e_t

Figure 3-2. Error of the four groups over time, where group three is shown in Figure 3-2(c)

Below is a row of three panels. It has no label, so it cannot be referenced, but it still takes a figure number:



(a) Option one

(b) Option two

(c) Option three

Figure 3-3. Results of the three options

This is a quotation: simulated data cannot fully represent real conditions, so these conclusions need further validation on real data.

4 Conclusion

We proposed an attention-based method for predicting sleep rhythm shifts. The experiments in Section 3 show that it outperforms the baselines. Future work includes:

- validation on real data;
- a study of the model's interpretability.

References

[1] Vaswani A, Shazeer N, Parmar N, et al. Attention is all you need[C]. Advances in Neural Information Processing Systems, 2017.

[2] Smith A. A made-up reference for testing how references look[J]. Journal of Examples, 2026.

Acknowledgements

Thanks to everyone who helped test this template. Like "References", this chapter is not numbered.

Appendix A - Survey Details

Appendices are lettered. See Appendix A.1, Table A-1, Figure A-1, and Eq. (A-1) below.

A.1 Questions

Table A-1. Survey items

Item	Scale
Sleep quality	1 to 5
Daytime fatigue	1 to 5

A.1.1 Scoring

The score is computed as follows:

$$S = \sum_{i=1}^{n} w_i x_i \tag{A-1}$$

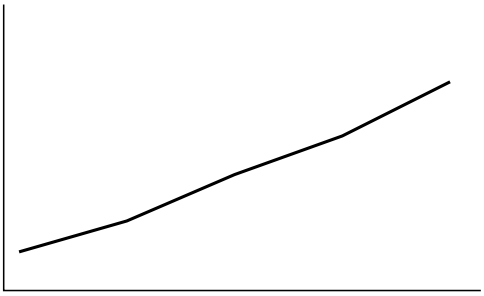


Figure A-1. A figure in the first appendix

A.1.1.1 Weights

This heading should be A.1.1.1. The scoring code is in Listing A-1:

Listing A-1. score.py

```
def score(weights, items):
    return sum(w * x for w, x in zip(weights, items))
```

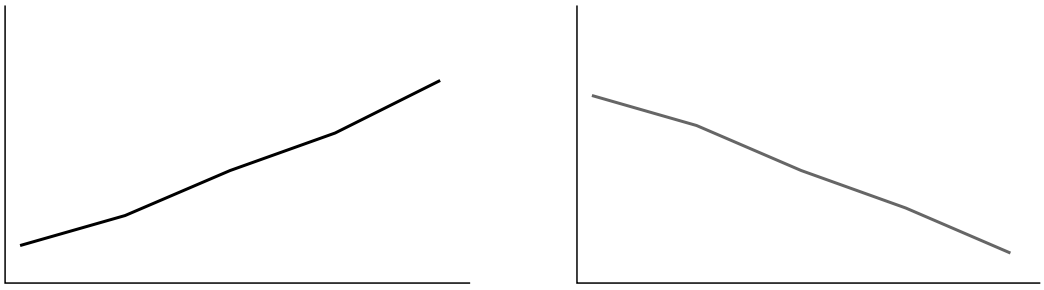
A.2 Notes

This heading should be A.2.

Appendix B

The second appendix should be B, and its numbering starts again. See Figure B-1, Figure B-1(a), and Appendix B.

B.1 Raw Data



(a) First panel

(b) Second panel

Figure B-1. A figure group in the second appendix

A wrong reference should stand out: **[@fig:missing]**.

Appendix C

```
#!/usr/bin/env python3
"""本地 Markdown 编辑器，可以打开任意文件夹。

用法：bash start.sh [文件夹]，或 python editor.py [端口]，然后打开
http://localhost:8001

只监听本机，并拒绝来自其他网页的请求，否则你打开的任意网站都可能改写你的文件。
要打开哪个文件夹由每个请求的 root 参数决定，服务端不保存“当前文件夹”，所以多个窗口可以
各开一个文件夹。
"""
import base64
import json
import mimetypes
import os
import posixpath
import re
import shutil
import subprocess
import sys
import tempfile
import threading
from datetime import datetime
from http.server import BaseHTTPRequestHandler, ThreadingHTTPServer
from pathlib import Path
from urllib.parse import parse_qs, quote, unquote, urlsplit

import yaml
from markdown_it import MarkdownIt
from mdit_py_plugins.anchors import anchors_plugin
from mdit_py_plugins.dollarmath import dollarmath_plugin
from mdit_py_plugins.footnote import footnote_plugin
from mdit_py_plugins.tasklists import tasklists_plugin
from pygments import highlight as pygments_highlight
from pygments.formatters import HtmlFormatter
from pygments.lexers import get_lexer_by_name
from pygments.util import ClassNotFound

import academic
import report_style

try:
```

```
    from send2trash import TrashPermissionError, send2trash
except ImportError:  # 只有删除文件需要它，没装时其他功能照常可用
    send2trash = None
    TrashPermissionError = OSError

DEFAULT_PORT = 8001
APP_DIR = Path(__file__).resolve().parent
STATIC_DIR = APP_DIR / "static"
RECENT_FILE = APP_DIR / "recent.json"
PREFERENCES_FILE = APP_DIR / "preferences.json"
STATIC_FILES = {
    "/": ("index.html", "text/html; charset=utf-8"),
    "/editor.js": ("editor.js", "text/javascript; charset=utf-8"),
    "/editor.css": ("editor.css", "text/css; charset=utf-8"),
    "/document.css": ("document.css", "text/css; charset=utf-8"),
}
MARKDOWN_SUFFIXES = (".md", ".markdown")
SKIP_DIRS = {"node_modules", "__pycache__", "venv"}  # 依赖和缓存，里面的
Markdown 不是你写的
MAX_TREE_ENTRIES = 50000  # 不小心打开整个硬盘时，不至于一直卡住
MAX_RECENT = 8
IMAGE_TYPES = {"image/png": ".png", "image/jpeg": ".jpg", "image/gif": ".gif",
"image/webp": ".webp"}
MAX_IMAGE_BYTES = 50 * 1024 * 1024
CODE_CSS = HtmlFormatter(style="default").get_style_defs("pre")
CDN_CODEMIRROR = "https://cdn.jsdelivr.net/npm/codemirror@5.65.16/"
CDN_KATEX = "https://cdn.jsdelivr.net/npm/katex@0.16.11/"
# 第二道防线：就算清理漏了，也不能运行内联脚本，脚本只能来自这些地址，也不能向其他网站发
请求
PAGE_SECURITY_POLICY = (
    "default-src 'none'; "
    f"script-src 'self' {CDN_CODEMIRROR} {CDN_KATEX}; "
    f"style-src 'self' 'unsafe-inline' {CDN_CODEMIRROR} {CDN_KATEX}; "
    f"font-src {CDN_KATEX} data:; "
    "img-src * data: blob:; connect-src 'self'; frame-src https:; "
    "base-uri 'none'; form-action 'none'; object-src 'none'"
)
FILE_SECURITY_POLICY = "default-src 'none'; style-src 'unsafe-inline'; sandbox"
# 直接打开 SVG 时，里面的脚本不能运行

recent_lock = threading.Lock()
```

```
class ApiError(Exception):
    def __init__(self, status, message):
        super().__init__(message)
        self.status = status
        self.message = message


# ---------- 路径 ----------
def workspace(root):
    """用户打开的文件夹，必须是绝对路径。"""
    path = Path(root) if isinstance(root, str) and root else None
    if path is None or not path.is_absolute():
        raise ApiError(400, "Enter the full path of a folder")
    if not path.is_dir():
        raise ApiError(404, f"Folder not found: {root}")
    return path.resolve()


def content_path(root, rel):
    """把相对路径转成文件路径，不允许离开打开的文件夹，也不允许碰 .git 这样的隐藏文
件。"""
    base = workspace(root)
    parts = rel.split("/") if isinstance(rel, str) else []
    invalid = not parts or "\\" in rel or ":" in rel or any(
        not part or part.startswith(".") or part.endswith((" ", ".")) for part
in parts
    )
```

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1. This is a footnote, used to test the font size and position of footnotes. ↩